

1) $Y \mid \alpha, \beta \sim \text{Poisson}(\lambda)$

$$\lambda = \exp(\alpha + \beta x)$$

$$\alpha, \beta \sim N(0, 10^2)$$

Use Metropolis Algorithm:

1) Initialize $\alpha^{(0)}, \beta^{(0)}$

2) take candidate distributions as
 $\alpha_c^{(i)} \sim N(\alpha^{(i)}, \sigma_\alpha^2)$ & $\beta_c^{(i)} \sim N(\beta^{(i)}, \sigma_\beta^2)$

3) Take $R = \min\left\{1, \frac{p(\alpha_c^{(i)}, \beta_c^{(i)} \mid Y)}{p(\alpha^{(i)}, \beta^{(i)} \mid Y)}\right\}$

3) Take $R = \min\left\{1, \frac{p(\alpha_c^{(i)} \mid \beta^{(i)}, Y)}{p(\alpha^{(i)} \mid \beta^{(i)}, Y)}\right\}$

& update α & β with probability R

4) Do the same for β

5) Repeat (2)-(4) until convergence & the required no of samples achieved.

2) a) $Y: 1, 0 \quad \text{iid } f(y|0) = \frac{1}{2} \{ \phi(y-0) + \phi(y) \}$

where $0 \in \mathbb{R}$

& $\phi(z) = \exp(-z^2/2) / \sqrt{2\pi}$

Let

$$f(Y, B) = g(Y|B) \cdot h(B)$$

where

$$g(X|B) = \phi(X - B \cdot 0)$$

$h(B)$ is Bernoulli pmf

$$f(x) = \sum_{b=0}^1 g(x|b) h(b)$$

$$= \phi(x) \frac{1}{2} + \phi(x-0) \frac{1}{2}$$

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4) let Y_{1i} denote the placebo observations
& Y_{2j} denote the treatment observations

where, $i = 1, \dots, m_1, m_1 = 6$
 $j = 1, \dots, m_2, m_2 = 6$

~~let~~ we use a two sample t-test
to test the effectiveness of the treatment

let's say

$$Y_{1i} \sim N(\mu, \sigma^2)$$

$$Y_{2j} \sim N(\mu + \delta, \sigma^2)$$

we want to test the hypothesis

$$H_0: \delta = 0$$

The marginal posterior of δ is given by

$$\delta | Y \sim t_m(\bar{Y}_2 - \bar{Y}_1, \frac{\hat{\sigma}^2}{m_1} + \frac{\hat{\sigma}^2}{m_2})$$

$$\bar{Y}_1 = \sum_{i=1}^{m_1} Y_{1i} / m_1, \quad \bar{Y}_2 = \sum_{j=1}^{m_2} Y_{2j} / m_2$$

$$\hat{\sigma}^2 = \left[\sum_{i=1}^{m_1} (Y_{1i} - \bar{Y}_1)^2 + \sum_{j=1}^{m_2} (Y_{2j} - \bar{Y}_2)^2 \right] / m$$

$$m = m_1 + m_2$$